University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Digital Logic Design	**Course Code:** CMPE-221
Assignment Type: Lab	**Dated:** 06th October 2026
Semester: 3rd	**Session:** 2025
Lab Number: 3	**CLOs to be covered:** CLO 1, CLO 2, CLO 3
Lab Title: Multiplexers, demultiplexers/decoders	**Teacher Name:** Fareed Jafri

Lab Goals

- Test the operation of an 8 line data input multiplexer (8x1 MUX), (CLO1, CLO3).
- Test the operation of an 8 line data output active low demultiplexer (3x8 DEMUX), (CLO1, CLO3).
- Making a logical circuit using an 8 line active low decoder (3x8 DEC), (CLO1, CLO2, CLO3).

Equipment Required

- **ICs:** 7408 (1x), 74138 (1x), 74151 (1x)
- **Tools:** IC puller (1x), Nose pliers (1x), Wire stripper (1x)

Lab Evaluation

	CLOs	Maximum marks	Marks obtained
Task 1a	1	2	
Task 1b	3	1	
Task 2a	1	2	
Task 2b	3	1	
Task 3a	2	1	
Task 3b	1	2	
Task 3c	3	1	
Total			**/10**

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric	CLOs
Completed	**100%**	L1	Completed the task correctly.	All
Great attempt	**75%**	L2	Almost completed the task, with minor deficiencies	2,3
Good attempt	**50%**	L3	Did not complete the task but managed to reasonably attempt it.	All
Basic attempt	**25%**	L4	Did not complete the task but managed to basically attempt it.	All
Did not attempt	**0%**	L5	Did not complete the task or attempt to do it.	All

Test the operation of an 8 line data input multiplexer (8x1 MUX), (CLO1, CLO3)

Task 1a [CLO1, 2 Marks]

The 74151 IC is is an 8 data input line multiplexer (8x1 MUX). The pin out of the IC is shown in the 'Pin out' section at the end. Wire up the IC as follows. Switches S7, S8 and S9 are connected to the IC's selction inputs S_2, S_1 and S_0 (to select the data input to the multiplexer). Data data inputs I_0 and I_1 are connected to ground. Data inputs I_2 and I_3 are connected to +5V. Switches S0 to S3 control the input variables w,x,y, and z, which are applied to the data inputs I_4 to I_7 respectively of the multiplexer. Connect E to 0V. Connect the output Z of the multiplexer to any LED of your choice. Show the instructor that the circuit works correctly.

Task 1b [CLO3, 1 Mark]

Turn the switches S0 and S1 OFF. Turn the switches S2 and S3 ON. Now using your circuit from task 1a fill in the following table.

Binary input $S_2S_1S_0$ (Switches S7, S8, S9)	**Binary output Z**
000	
001	
010	
011	
100	
101	
110	
111	

Test the operation of an 8 line data output active low demultiplexer (3x8 DEMUX)

Task 2a [CLO1, 2 Marks]

The 74138 IC is an 8 data output line active low demultiplexer (3x8 DEMUX). The pin out of the IC is shown in the 'Pin out' section at the end. Wire up the IC as follows. Switch S7, S8 and S9 are connected to the IC's selection inputs C, B and A (to select the data output line of the demultiplexer). Switch S0 controls the data input ($G_{2A)}$. The enable inputs G_{2B} and G_1 have to be connected to 0V and 5V respectively. Data outputs Y_0 to Y_7 are connected to LEDs L0 to L7 respectively. Show the instructor that the circuit works correctly.

Task 2b [CLO3, 1 Mark]

Now using your circuit from task 2a fill in the following table.

Binary input CBA	**Value of input from S0**	**Binary output $Y_0Y_1Y_2Y_3Y_4Y_5Y_6Y_7$**
000	0	
000	1	
001	0	
001	1	
010	0	
010	1	
011	0	
011	1	
100	0	
100	1	
101	0	
101	1	
110	0	
110	1	
111	0	
111	1	

Making a logical circuit using an 8 line active low decoder (3x8 DEC)

Task 3a [CLO2, 1 Mark]

Show how a 4-input AND gate with inputs w, x, y and z, and output F can be made using 2-input AND gates. Draw your solution in the space provided below.

Task 3b [CLO1, 2 Marks]

The pin out of the 7408 IC is shown in the 'Pin out' section at the end. Make the 4-input AND gate of task 3a using this IC, then connect the outputs Y_0,Y_1,Y_4,Y_5 of the circuit in Task 2a to the inputs (w, x, y and z) of this AND gate. Connect the output (F) of the AND gate to LED L7. **Turn S0 OFF, this will convert the demultiplexer into a decoder**. Now show this circuit to the instructor.

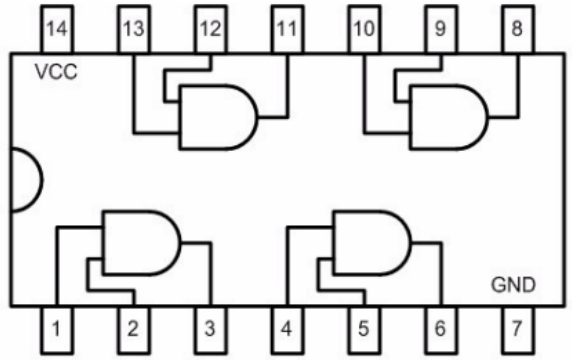
Task 3c [CLO3, 1 Mark]

Using your circuit from task 3b fill in the following table.

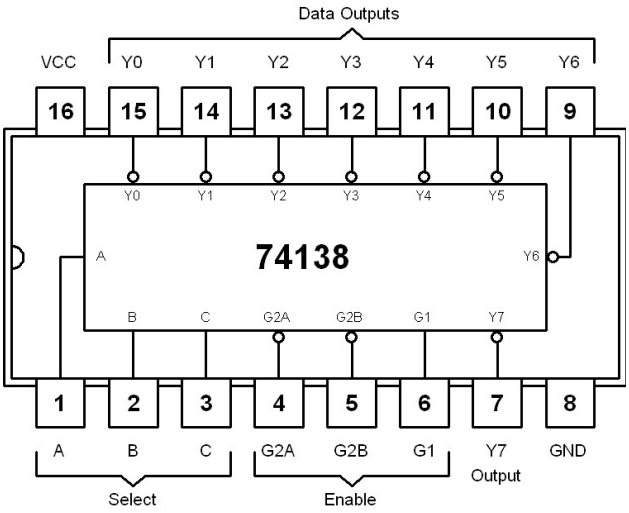
Binary input CBA	**Binary output of AND gate F**
000	
001	
010	
011	
100	
101	
110	
111	

Pin out

7408 IC



74138 IC



74151 IC

